Title

Readme for the replication codes

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1 Hardware and software requirements (including packages and libraries) and the expected running time

1.1 Hardware requirements

The code was run on a Dell XPS 15 9550 computer with the following specifications:

- operating system: Windows 10 Pro (64-bit) (10.0, Build 19045)
- processor: Intel Core i7-6700HQ CPU @ 2.60GHz (8 CPUs)
- BIOS: 1.14.0 (type: UEFI)
- RAM: 16.00 GB

1.2 Software requirements

- Matlab (Version 9.3, Release 2017b)
  - Simulink: Version 9.0
  - Control System Toolbox: Version 10.3
  - Econometrics Toolbox: Version 4.1
  - Financial Instruments Toolbox: Version 2.6
  - Financial Toolbox: Version 5.10
  - Global Optimization Toolbox: Version 3.4.3
  - Optimization Toolbox: Version 8.0
  - Parallel Computing Toolbox: Version 6.11
  - Partial Differential Equation Toolbox: Version 2.5
  - Statistics and Machine Learning Toolbox: Version 11.2

1.3 Expected running time

The total runtime required is approximately 1 day.

2 Data

This section provides a comprehensive overview of the data used. It includes a Data Availability Statement detailing the source and accessibility of each dataset, followed by citations for data sources. Additionally, a summary of data descriptions and files is provided, outlining the structure and content of the datasets and file formats used for analysis and estimation.

2.1 Data availability statement

The data used in this replication package is sourced from the Federal Reserve Economic Data (FRED) managed by the Economic Research Division of the Federal Reserve Bank of St. Louis. The dataset is publicly accessible and can be obtained from the following link: https://fred.stlouisfed.org

2.2 Data citations

Federal Reserve Economic Data, Economic Research Division, Federal Reserve Bank of St. Louis. Available at: https://fred.stlouisfed.org

2.3 Data descriptions and files

Table 1: Data description

Variable	Description	Source	Sample	Transformation
Y_t	Macroeconomic variable	Source A	2000Q1–2020Q4	Log-transformed and demeaned.
R_t	Policy variable	Source B	2000Q1–2020Q4	Mean-adjusted.
P_t	Price index	Source C	2000Q1–2020Q4	Log difference.
W_t	Wage variable	Source D	2000Q1–2020Q4	No transformation.

Table 2: Data structure for variables and files

File	Description
`variable1.xls`	Data series for the price index.
`variable2.xls`	Data series for the wage.
`rawData.mat`	Raw data file containing the data in its initial form, before processing.
`fullsampleData.mat`	Full sample dataset used for illustration purposes.
`data.wf1`	Workfile containing original and transformed series.
`data.xls`	Spreadsheet containing the variables used for estimation.
`data.mat`	Final database file used for estimation after processing.

3 Description of programs

This section provides an overview of the replication process for the various figures and tables included in the analysis. For each replication step, the code used, the input data files, and the expected output files are documented in the table. The table also reports the appendix figures and tables for supplementary materials.

Table 3: Code and output structure

Step	Code and Output
Step 1: Illustration	• Figure generated by the code. • Placeholder for graphical output.
Step 2: Results	• Code: `codes_script.m` • Input: `input_data/`, `model_files/` • Output: `results_results.mat`

4 Instructions

1. **Set the Current Folder in MATLAB**: Open MATLAB and set the current working directory to the folder where this repository has been saved or extracted. You can do this by navigating to the folder in the MATLAB interface or by using the command:

   ```
   cd('path_to_repository')
   ```

   Replace `path_to_repository` with the actual path where the files are stored.

2. **Add Subfolders to the MATLAB Path**: The repository includes subfolders, such as `data`, `codes`, `results`, ensure they are added to the MATLAB path by running:

   ```
   addpath(genpath(pwd))
   ```

3. **Running the Main Script**: To reproduce the results, execute the following main script:

   ```
   run('replication.m')
   ```

   The script will load and use the necessary raw data, process it, and generate the results, in the `results` folder.

References

References here.